Jacob King

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SUMMARY

Self-taught engineer focused on AI reliability and evaluation, with two years of regulated records work under FERPA and HIPAA and five years running daily operations. Building since 2024: started with predictive models for economic time series, which became the signal layer of an autonomous trading system run against live markets, whose safety architecture was then generalized into an open-source runtime supervisor for AI agents. The through-line is the same problem in both careers: making sure the record matches what actually happened, and catching it early when it doesn't.

TECHNICAL PROJECTS

Agent Supervisor — open-source runtime supervision for AI agents

Python | 2026 | github.com/JakeKing-Dev/Agent-Supervisor

- **Built a zero-dependency Python library** that detects behavioral failure in autonomous AI agents from a partial execution trace — action loops, collapsing diversity, retry escalation, stalled state — and halts the run before cost accumulates. 27 unit tests; CI across Python 3.10–3.13.
- **Analyzed 1,408 human-annotated agent trajectories** (AgentRewardBench, four benchmarks) to identify which behavioral signals predict failure. Runs annotated as looping failed 93.7% of the time versus 49.7% for those that did not — the finding that motivated a mechanical detector running in microseconds rather than an LLM judge that cannot sit inline in production.
- **Designed fail-closed recovery semantics:** a halted agent may take only reducing actions, and resuming requires externally verified reconciliation evidence rather than self-reported health.
- **Instrumented and deployed it:** eight Prometheus metrics served from the standard library, Docker image, provisioned Grafana dashboard, and a Kubernetes sidecar manifest with liveness and readiness probes, resource limits and a read-only root filesystem.
- **Built the operator dashboard and its end-to-end test suite:** 15 Playwright tests covering live metric rendering, freeze state, failure of the metrics endpoint and small-screen layout, running in CI on every push. Writing them surfaced two defects in the metrics layer — a headline counter that was never fed, and halt events counted per verdict rather than per halt.

BugNet — completeness and reconciliation monitoring

Python | 2026 | github.com/JakeKing-Dev/bugnet

- **Built an open-source watchdog** that audits a system's records against an outside source of truth rather than against itself, and blocks the component at fault instead of raising an alert. Zero dependencies, 25 tests, CI across Python 3.10–3.13.
- **Published a measured detection benchmark:** injected dropped outcomes, unrecorded work and ledger drift at varying rates, then reported detection rate, time to detect and false positives — every injected fault caught, zero false positives across ten clean runs, with the reconciliation blind spot measured and documented rather than hidden.
- **Came from a real incident:** a system whose records showed a profit on a day the account fell, because losing outcomes were dropped before being recorded while winners were kept — survivorship bias inside the bookkeeping that also fed the evidence promoting strategies.

Autonomous Trading System — event-market trading bot

Python | 2024–2026 | private repository, available on request

- **Began as self-directed study in predictive modeling** of economic time series in 2024; that forecasting work became the signal layer of the system below.
- **Grew it into a production system** traded unsupervised against live markets with real capital, developed continuously across two years: a signed REST client, order state machine, persistence and crash recovery, risk gating, reconciliation and strategy promotion.
- **Implemented the risk and safety layer:** a pre-trade gate that fails closed on any pricing or safety fault, fractional-Kelly position sizing, exposure and loss caps, and exact decimal fee arithmetic.
- **Built two-pass reconciliation against the exchange** plus a freeze manager whose recovery path verifies reconciliation freshness independently rather than trusting a caller-supplied status.
- **Designed a shadow-mode promotion ladder:** new strategies traded simulated capital at live prices and had to earn a measured track record before being granted real money, with automatic demotion when live results lagged.

TECHNICAL SKILLS

- **Languages:** Python (primary), Bash, JavaScript, HTML/CSS
- **Testing and QA:** Playwright end-to-end browser testing, unit testing (unittest), test design around failure modes and regression cases, CI pipelines in GitHub Actions, defect isolation and reproduction
- **Infrastructure:** Docker, Docker Compose, Kubernetes manifests, Prometheus, Grafana, Git and GitHub, Linux command line
- **Data:** time-series analysis, statistical calibration, API integration
- **AI:** agent evaluation and failure analysis, trajectory analysis, LLM-as-judge limitations, AI-assisted development
- **Practice:** automation scripting, fail-closed system design, technical documentation, Windows and general IT troubleshooting

PROFESSIONAL EXPERIENCE

Paraprofessional

Frank Zeek Elementary School — Ukiah, CA | August 2026 – Present

- Maintain individualized student support records under federal confidentiality requirements, with daily tracking and documentation audited for compliance.
- Coordinate with teachers, administrators and specialist providers to audit progress logs and reconcile official compliance files.

Paraprofessional

Oak Manor Elementary School — Ukiah, CA | October 2025 – June 2026

- Reviewed and maintained confidential Individualized Education Plans and student records in full compliance with FERPA and HIPAA requirements.
- Monitored multi-track schedules and verified documentation accuracy for progress reports and administrative review.

Paraprofessional

Willits Elementary Charter — Willits, CA | August 2024 – June 2025

- Managed and updated sensitive educational records under federal and state confidentiality regulations, tracking compliance against individualized plans.
- Prepared and organized complex documentation for administrative review and distribution.

Theatre Manager

Noyo Theatre — Willits, CA | June 2017 – May 2022

- Ran daily operations for a single-site venue: scheduling, cash handling and nightly register reconciliation against recorded sales.
- Performed inventory audits and maintained operational and financial logs.
- Developed staff training and monitored task completion across shifts.

EDUCATION

- **Associate of Arts, Political Science (Pre-Law)** — Mendocino College. In progress, 4.0 GPA.
- **High School Diploma** — Willits High School, 2017.